

SDP



GitHub Copilot ACP

Your Tool Just Got an AI Agent

One flag. Any editor. Full agent capabilities.

"I build tools outside VS Code — how do I give them Copilot's full agent capabilities?"

Every role needs something different: real integration speed, enterprise trust, or multi-agent coordination.

ACP gives each of you exactly what you need.

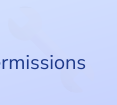
Zed Developer

Connect without months of custom work



Platform Engineer

Ship safely with policy-enforced permissions



Orchestrator

Coordinate agents across repositories



2 weeks

vs 3 months to ship before ACP



14 lines

of TypeScript to connect



3 lines

of policy to secure automation

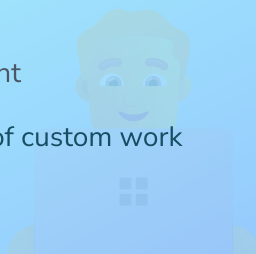


PART 1

Zed Developer

From locked-in to full agent

Connect without months of custom work



PART 2

Platform Engineer

Three permission strategies

Ship safely with policy-enforced permissions

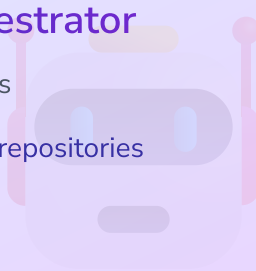


PART 3

Multi-Agent Orchestrator

One protocol, many agents

Coordinate agents across repositories



PART 4

Ship Safely

When and how to use ACP

Decision framework for adoption



Click any section to jump directly there

PART 1

Zed Developer

From locked out to full agent experience in 14 lines of code



One Command

`copilot --acp --stdio`



Streaming Responses

Real-time agent output



Any Editor

Standard protocol for all

Connect any editor to Copilot

↳ 2 weeks instead of 3 months

From Command to Live Agent Workflow

acp-protocol

```
→ {method: initialize, params: { ... }}
← {result: initialized}

→ {method: prompt, text: Refactor fetch}
← {event: thinking, text: Analyzing ... }
← {event: tool, tool: read_files, path: *.ts}
← {event: thinking, text: Found 3 fetches}
← {event: text, delta: function getUsers() {
  return Promise.all([
    fetch(/api/users),
    fetch(/api/posts)
  ])
}}
}}
```



Streaming protocol

Real JSON-RPC messages: → client, ← server



Visible agent internals

See thinking, tool calls, and code as it streams



Language-agnostic

Any editor can implement this protocol

From Locked-In to Free

Before ACP

Stuck waiting for IDE team

3+ months

Build custom agent API

Limited to existing features

After ACP

Connect in 2 weeks

Standard protocol, any tool

Full agent reasoning available

Your control

No waiting for platform updates

90%

less integration code

2-3 weeks

time to market

∞

use cases

PART 2

Platform Engineer

Three permission strategies prove ACP is production-safe



Interactive Approval

Human reviews each request



Policy-Based

Auto-approve safe ops



Tiered by Env

Different rules per context

```
Permission callback gates every action  
↳ 3 lines to auto-approve reads, prompt writes
```

Permissions Done Right

✖ The Challenge

Ship agents to production safely

Balance:

Power vs. policy enforcement

💡 The Approach

Permission callback intercepts requests

Three strategies:

Interactive, Policy-based, Tiered by env

✅ The Result

Auditable per-session permissions

Your rules.

Enforced at the agent boundary

Permission Policies That Scale

policy.ts

```
const SAFE = [read, search];
const BLOCKED = [delete];

async requestPermission(p) {
  if (SAFE.includes(p.tool))
    return approved;
  if (BLOCKED.includes(p.tool))
    return cancelled;
  return await promptUser(p);
}
```



Auto-approve safe ops

Reads never prompt



Auto-block dangerous ops

Deletes always blocked



Human decides the rest

Writes prompt the user

PART 3

Multi-Agent Orchestrator

One protocol, many agents, coordinated work across repos



Broadcast

Prompt all workers at once



Synthesize

Orchestrator merges results



Compose

Standard sessions, no special mode

Spawn one agent per repo
↳ Coordinate results naturally

Spawning Multi-Agent Sessions

orchestrator.ts

```
const workers = repos.map(repo => ({
  process: spawn(copilot, [--acp, --stdio]),
  repo
}));

for (const worker of workers) {
  const sess = await connection.newSession({
    cwd: worker.repo.path,
    mcpServers: [filesystemServer(repo.path)]
  });
}
```



One process per repo

Isolated context and state



Same protocol everywhere

No special multi-agent mode



Compose at will

Standard ACP sessions compose naturally

Five Scenarios, One Operator Playbook

SCENARIO.md

1. service-audit-and-api-alignment
2. payment-expansion
3. loyalty-feature-rollout
4. inventory-sync-coordination
5. security-audit-shared-upgrade

Suggested: 1 → 2 → 3 → 4 → 5

🔍 Start with architecture truth

Scenario 1 creates a shared baseline across all repos

🔄 Escalate complexity

Scenarios 2-4 layer feature rollout and dependency routing

🚨 End with critical path ops

Scenario 5 proves security sequencing under pressure



Scenario 1: Broadcast First, Synthesize Second

The fastest way to align eight shopfleet services

```
$ Broadcast to all workers
> "Audit this microservice: purpose, APIs, dependencies, risks"
🤖 Workers streaming:
  users: profile + auth ownership
  orders: fulfillment + payment handoff
  storefront: UI + checkout orchestration

$ Set Orchestrator Focus
> "Synthesize dependency map + alignment gaps + next 3 actions"
✅ One architecture narrative from seven isolated contexts
```

Operator pattern: shared prompt -> worker evidence -> orchestrator synthesis

Scenario 3: Loyalty Rollout with Cross-Repo Cascades

acp-manifest.json

```
{
  "name": "shopfleet-storefront",
  "dependsOn": ["shopfleet-users", "shopfleet-products"]
}

// Broadcast: implement loyalty points
// Route plan: storefront ← users, products, payments
// Cascade: service-specific prompts generated per dependenc
```

Graph-aware prompts

Routes are generated from repo relationships, not guesswork

Repo-specific execution

Each worker gets the same intent but tailored implementation context

Consistency checks

Orchestrator catches schema drift across LoyaltyAccount and PointsTransaction

Scenario 5: Shared Library Security Upgrade

Order of operations is the difference between safety and outage

✗ Ad-hoc rollout

- 1 Patch services in parallel
Inconsistent hash behavior
- 2 Discover breakages after deploy
Login failures surface late
- 3 No migration plan
Users forced into resets
- 4 Manual coordination
High incident risk



Uncontrolled security blast radius

✓ Orchestrated rollout

- 1 Patch shopfleet-shared first
Single source of crypto truth
- 2 Broadcast diagnostic audit
Find all hash usage before rollout
- 3 Dual-verify transition
Old SHA + new bcrypt during migration
- 4 Orchestrator validates all services
Evidence before production



Controlled migration with audit trail

Critical path: shared library -> dependent services -> validation broadcast

Live Screenshot Storyboard for Your Polyrepo Demo

1

Topology

Full app layout: orchestrator + 3+ workers

Dependency graph with node and edge counts

Banner showing unloaded dependency neighbors

2

Coordination

Routing plan panel with edited downstream prompts

Broadcast results with coalesced worker outputs

Orchestrator card in Synthesizing state

3

Agent States

Worker card with dependsOn and dependedBy pills

Unloaded dep chip with Load as Worker action

Session panel showing Restore vs Re-spawn

📸 If you want backup visuals, cli-acp has walkthrough captures under docs/images/walkthrough/01-06 while you replace with live local shots.

PART 4

Ship Safely

Decision table and alternatives: When to use ACP, when not to



Use ACP

Non-VS Code, multi-agent, CI/CD



Maybe Not

Already in VS Code is simpler



Choose Wisely

Know your alternative

Four personas, one next step each
↳ Flywheel: more clients → more agents

Choose Your Path



Use ACP Now

Non-VS Code tool, multi-agent coordination, Claude API, enterprise policy



Consider Alternatives

Already VS Code, IDE-specific needs, single-agent only, built-in copilot works



Start Small

Read spec, try SDK for your language, deploy to dev/test, add policies before prod

✓ What You Can Do Today

Today

- Follow agentclientprotocol.com/spec
- Bookmark the official documentation
- Star the SDK repositories

This Week

- Clone the ACP SDK for your language
- Run the 14-line example and stream responses
- Read the four-layer architecture guide

This Month

- Integrate ACP into your editor or tool
- Test permission strategies in dev/stage
- Merge to production with policy gates

🔑 Key Takeaway

One flag. One agent. Ship safely with ACP.

Official Documentation

[ACP Protocol Overview](#)

Complete specification

[Copilot CLI ACP Server](#)

Integration reference

[Four-Layer Architecture](#)

Design deep dive

Code & Examples

[ACP SDKs & Specs](#)

TypeScript, Python, Rust, Kotlin

[Orchestrator Reference](#)

Multi-agent patterns



Your Tool Just Got an AI Agent

Build with ACP. Ship safely. Scale.



Start here: github.com/agentclientprotocol/spec